Sankhadip SAHA

Aqueous Zinc-Ion Batteries | Electrochemistry | Computational Modelling | Practical Battery Systems

Contact

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Research Profile

Electrochemical energy storage researcher focused on uncovering the mechanisms governing performance, degradation, and scale-up in aqueous energy storage systems. My research integrates operando characterisation, electrochemical analysis and computational modelling to resolve charge-storage pathways in Mn-based cathodes and understand how mechanical constraints influence electrochemical behaviour in practical pouch-cell architecture. By bridging fundamental mechanisms to device-level operation, I aim to establish design principles for safe, durable, and scalable batteries.

Research Highlights

- Uncovered the electrochemical mechanism governing Mn-based Prussian Blue Analogue (PBA) cathodes in Aqueous Zinc-Ion Batteries (AZIBs) through operando XRD, ex-situ XANES and first-principles modelling, revealing how framework reconstruction and Mn dissolution-redeposition underpin the apparent stability frequently reported at high current densities.
- Elucidated proton-primed dissolution-deposition pathways as a central charge-storage mechanism in MnO_2 cathodes through the integration of operando analyses, electrochemical investigations, density functional theory (DFT), and ab-initio molecular dynamics (AI-MD).
- Revealed the critical role of stack pressure in governing gas evolution, interfacial contact, and zinc anode stability through pressure-controlled AZIB pouch cell platforms.
- Developed pressure-controlled AZIB platforms to elucidate how mechanical constraints govern electrochemical behaviour across intercalation (Zn||ZVO), conversion (Zn||I_2) and dissolution-deposition (Zn||MnO_2) cathode chemistries at pouch cell scale.
- Integrated operando neutron imaging, ultra-small angle scattering and X-ray CT to elucidate gas evolution and spatial distribution in aqueous Zn-ion pouch cells and quantify cross-sectional porosity evolution in cycled Zn electrodes

Education

- **University of New South Wales (UNSW)** — *Sydney, Australia*
  Doctor of Philosophy (PhD) — *May 2023-Nov 2026*
  Chemical Engineering — ***(Thesis submission in Nov 2026)***
  ***Thesis title**: "Design Principles for Aqueous Zinc-Ion Batteries: From Fundamental Mechanisms to Practical Pouch-Cell Design"*
- **Indian Institute of Technology, Bombay (I.I.T. Bombay)** — *Mumbai, India*
  Master of Technology (MTech) — *August 2019-August 2021*
  Metallurgical Engineering and Materials Science
  ***Thesis title**: "Investigating the Interaction of Biomolecular Systems with Graphene-like Materials"*
  (Grade: 8.87/10)
- **Government College of Engineering and Ceramic Technology** — *Kolkata, India*
  Bachelor of Technology (BTech) — *August 2014-July 2018*
  Ceramic Technology
  ***Thesis title**: "Structural and Adsorption Characteristics of Nano-Crystalline NiO"*
  (Grade: 8.80/10)

Research and Professional Experience

- **Project Associate-I (Junior Research Fellow)** — *Oct 2021-March 2023*
  CSIR-Central Glass and Ceramic Research Institute (CGCRI) — *Kolkata, India*
  - Designed ZnO-incorporated hydrogel biomaterials, establishing their antibacterial, cytocompatible and wound healing efficacy.

- Developed marine sponge-derived silica/PCL electrospun nanofibrous scaffolds that promoted osteoblast adhesion and cytocompatibility for bone tissue engineering.

- **Summer Intern (Remote)** *June 2020-August 2020*
  Houston Community College South
  - Applied computational modelling to elucidate the inhibition mechanisms of organic molecules on mild steel corrosion, enabling the rational design of improved corrosion inhibitors.

- **Graduate Engineer Trainee (GET)** *July 2018-July 2019*
  Cera Sanitaryware Limited *Gujarat, India*
  - Revised manufacturing Standard Operating Procedures (SOPs) for 104 products, improving process standardisation and quality assurance

- **Summer Internships** *Jun 2016-Jul 2016, Jan 2017- Mar 2017, May 2017-Aug 2017*
  Indian Institute of Technology, (B.H.U.) *Varanasi, India*
  Department of Ceramic Engineering
  - Investigated V_2O_5-assisted liquid-phase sintering of Mn-doped $CaCu_3Ti_4O_{12}$ ceramics, reducing the sintering temperature by 150ºC while improving dielectric performance
  - Developed microwave-sintered $BaTiO_3$-$SrTiO_3$ (BTS) ceramics, achieving high dielectric constants with significantly reduced processing time via microwave sintering
  - Optimised V_2O_5-assisted liquid-phase sintering of $BaTiO_3$-$SrTiO_3$ (BTS) ceramics, lowering sintering temperature by 250ºC and enhancing structural properties.

Teaching Experiences

- **Teaching Assistant (TA)** *Sep 2023-April 2025*
  University of New South Wales (UNSW) *Sydney, Australia*
  - Courses: Process Modelling and Analysis (CEIC3000), Entrepreneurship and the Innovation Cycle (CEIC8204), Advanced Food Engineering (CEIC8450)
  - Contributed to courses through tutorials, examination materials, invigilation, grading and evaluation.

- **Teaching Assistant (TA)** *Aug 2019-July 2021*
  Indian Institute of Technology, Bombay (I.I.T. Bombay) *Mumbai, India*
  - Courses: Electronic Properties of Materials (MM318), XRD Lab RA

Experimental and Computational Skills

- **Electrochemical Techniques**
  - **Battery Fabrication:** Swagelok, coin, and pouch cells
  - **Electrochemical testing:** Galvanostatic Cycling, Cyclic Voltammetry (CV), Electrochemical Impedance Spectroscopy (EIS), Galvanostatic Intermittent Titration Techniques (GITT), Distribution of Relaxation Time (DRT)
  - **Operando electrochemistry:** Pressure-controlled pouch cell testing
  - **Instrumentation:** Landt battery cycler, Biologic VMP3 potentiostat, Palmsens potentiostat, Loadstar pressure sensor, Tekscan I-Scan Pressure Mapping, Electrode Die Cutter, Pouch Cell Vacuum Sealer, Electrode Calendaring Machine.
- **Materials Processing and Characterisation**
  - **Advanced:** Solution-route and Solid-State Synthesis, Operando XRD, SEM-EDS, Laser Optical Microscopy, FT-IR, TGA, DTA, UV-Vis, Raman, Rietveld Refinement.
  - **Basics:** XANES, Neutron Imaging, X-ray CT, TEM, ICP-OES, CHNS, Glovebox operation
- **Computational Materials Modelling**
  - **Advanced:** DFT, AI-MD, NEB, Bader Charge Analysis, Electronic Band Structure, COHP, Adsorption Energy, Formation Energy, Intercalation Voltage calculations
- **Software**
  - **Advanced:** VASP, Quantum Espresso, ImajeJ, X'Pert HighScore, GSAS-II, Full Prof Suite LOBSTER, VASPKIT, ASE, Python, VESTA, XCrysden, VMD, Origin, SasView.
  - **Basics:** DragonFly, Gatan DigitalMicrograph, IgorPro

Awards, Scholarships and Grants

- Selected to attend the ANSTO-HZB-AONSA Neutron School 2026 to be held at ANSTO, Sydney
- Secured Beamtime at ANSTO through the ACNS & NDF Merit Access Proposal (2026)
- Selected to attend the X-Ray & Neutron Imaging Workshop 2026 organised by ANSTO in Melbourne
- Secured the UNSW Graduate Research School Development and Research Training Grant (DRTG) for attending conferences (2026)
- Received the Professional Development Grant from the School of Chemical Engineering, UNSW, for attending conferences (2025)
- University International Postgraduate Award for pursuing doctoral studies at UNSW, Sydney (2023-2026)
- Ministry of Human Resources Development, Government of India, for postgraduate studies at IIT Bombay (2019-2021)
- 2nd place in poster presentation at International Conference on Global Trends in Traditional to Space Ceramics (GT-TSC), IIT-BHU Campus, Varanasi, India, 2022.

Journal Articles

Total articles published: 9, Citations: 75, h-index: 5, i10-index: 1 [As per Google Scholar upto October 2026]

First Author:

1. ***S. Saha**, Y. Shang, R. Ma, X. Zheng, P. V. Kumar, D. Kundu, "Unmasking the Intrinsic Mechanistic Limits of Manganese Prussian Blue Analogues in Aqueous Zn-Ion Batteries.", **Chemical Science**, 2026, https://doi.org/10.1039/D6SC02408D*
2. *B. Roondhe[+], **S. Saha[+]**, W. Luo, R. Ahuja, S. Saxena, "Detection of type-II diabetes using graphene-based biosensors.", **Journal of Physics D: Applied Physics**, 2024, 57(18), 185402. https://doi.org/10.1088/1361-6463/ad2336 **([+]co-first author)***
3. ***S. Saha**, S. Roy, S. Jana, T. Sarkar, S. Chanda, P. Datta, B. Chakraborty, S. Mondal, S. Halder, B. Kundu, S. K. Nandi, "Zinc Oxide Nanoparticles Incorporated in Poly-Hydroxyethyl Methacrylate/Acrylamide Membrane Trigger the Key Events of Full-Thickness Wound Healing in a Rabbit Model.", **Journal of Biomedical Materials Research Part B: Applied Biomaterials**, 2024, 112(12), e35510. https://doi.org/10.1002/jbm.b.35510*

(**2 first-author** manuscripts from my PhD work on pressure-controlled pouch cell platform for AZIBs are currently being drafted)

Co-author:

4. *Y. Shang, **S. Saha**, H. Wen, Q. Zhang, X. Wu, B. Hoex, M. Wang, N. Wang, T. Luo, S. Purohit, G. S. Gautam, W. M. Dose, L. Thomsen, S. Chang, P. V. Kumar, D. Kundu, "Decoding MnO2 redox chemistry from mechanistic ambiguity to design principles for aqueous Zn-ion batteries.", **Nature Communications**, 2026. https://doi.org/10.1038/s41467-026-74350-z*
5. *B. Xie, Y. F. Zhu, M. Shakeri, S. Jin, G. O'Connell, **S. Saha**, M. Mensi, P. V. Kumar, J. S. Luterbacher, E. C. Lovell, R. Amal, O. Kröcher, "Flame-Made Surface-Substituted Copper–Ceria as an Excellent Reverse Water–Gas Shift Reaction Catalyst via Three Reaction Pathways.", **Journal of the American Chemical Society**, 2025, 147(36), 32649–32661. https://doi.org/10.1021/jacs.5c07701*
6. *Z. R. Ramadhan, S. Cheong, **S. Saha**, Q. Li, X. Zheng, S. V. Somerville, A. R. Poerwoprajitno, P. V. Kumar, L. Dai, J. J. Gooding, R. D. Tilley, "Phase Boundary Engineering of Co_2 P-CoP Branched Nanoparticles Enhances Cobalt Oxidation for Oxygen Evolution Electrocatalysis.", **Advanced Materials**, 2026, e23118. https://doi.org/10.1002/adma.202523118*
7. *Y. Liu, R. Kokate, **S. Saha**, P. V. Kumar, "Rh, Pd, and Pt doping effects on the interaction between Ag20 nanoclusters and graphene oxide.", **Computational Materials Science**, 2026, 272, 114840. https://doi.org/10.1016/j.commatsci.2026.114840*
8. *M. Mani, K. Mariandry, U.V. Ghorpade, **S. Saha**, R. Kokate, R. Mishra, M. P. Nielsen, R. Tilley, B. Xie, M. P. Suryawanshi, P. V. Kumar, "Plasmonic Hot-Carrier Engineering at Bimetallic Nanoparticle/Semiconductor Interfaces: A Computational Perspective.", **Small**, 21(11), 2025, 2410173. https://doi.org/10.1002/smll.202410173*
9. *Y. Y. Ch'ng, **S. Saha**, M. Zhang, S. Zhou, L. Jiang, P. Ramadhany, J. Vongsvivut, A. Cernescu, P. V. Kumar, Y. Shan, Z. Ma, X. Luo, R. Daiyan, Z. Han, Rose Amal,"* Chlorine-Functionalized Silane Modified Copper Electrocatalyst for Enhanced CO_2 Reduction to Multi-carbon Products", ***Advanced Science**, e76741, https://doi.org/10.1002/advs.76741*

Conference Presentations

1. ***S. Saha (Poster Presenter)**, P. V. Kumar, D. Kundu, "Unravelling the Origin of Capacity Fading in Manganese-Based Prussian Blue Analogues for Aqueous Zinc Ion Batteries", 2nd Annual NSW Electrochemistry Symposium, Newcastle, NSW, Australia, 2025*

2. ***S. Saha (Poster Presenter)**, P. V. Kumar, D. Kundu, "Mechanistic Insights into the Electrochemical Behaviour of Manganese-Based Prussian Blue Analogues in Aqueous Dual-ion Batteries", International Conference on Nanomaterials and Atomaterials Science and Applications (ICNASA), RMIT University Campus, Melbourne, VIC, Australia, 2025*

3. ***S. Saha (Poster Presenter)**, H. Das, S. Sengupta, N. L. Thakur, B. Kundu, "Biocompatible Siliceous Converted Marine Sponge/PCL Electrospun Composite Fibres for Bone Tissue Regeneration", International Conference on Global Trends in Traditional to Space Ceramics (GT-TSC), IIT-BHU Campus, Varanasi, India, 2022.*

4. ***S. Saha (Speaker),** B. Roondhe, S. Saxena, "Investigating the interactions of Valine with Modified Graphene: A Dispersion Corrected Density Functional Theory Calculation", International Conference on Nanoscience and Nanotechnology (ICONN), SRM-IST, Virtual Conference, 2021.*

5. ***S. Saha (Poster Presenter)**, R. Bhattacharyya, T. K. Bhattacharyya, "Structural and Adsorption Characteristics of Nano-Crystalline NiO", International Conference on Expanding Horizons of Technological Applications of Ceramics and Glasses (EH-TACAG), College of Engineering (COEP), Pune, India*

Academic Mentoring Experience

- Mentored 3 Master's and 1 Bachelor's student in the school of Chemical Engineering, UNSW, providing research guidance, technical training and thesis supervision.
- Mentored undergraduate students on a reading project in Nanoscience and Technology for the Summer of Science (SoS) Program, organised by the Maths and Physics Club, I.I.T. Bombay.

Volunteer Experience

- Trained school students in the Student Solar Ambassador Workshop organised by SoULS, IIT Bombay, where students learn to assemble their Solar Study Lamp.

Languages

- **Full Professional** – English (proficiency test scores: TOEFL – 106/120)
- **Elementary** - German
- **Native** - Hindi and Bengali

References

- Prof. Priyank Vijaya Kumar
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  University of New South Wales (UNSW), Sydney
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- Prof. Dipan Kundu
  School of Chemical Engineering
  University of New South Wales (UNSW), Sydney
  Email ID: d.kundu@unsw.edu.au

- Prof. Sumit Saxena
  Department of Metallurgical Engineering and Materials Science, IIT Bombay
  Email ID: sumit.saxena@iitb.ac.in

- Prof. (Retired) Devendra Kumar
  Department of Ceramic Engineering,
  IIT(BHU) Varanasi
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